

Minimum Trading Risk Quantity

The minimum trading risk quantity is calculated as follows.

Let r be the current portfolio and let y_k indicate the number of shares to trade in stock k . Then these vectors are:

$$r = \begin{pmatrix} r_1 \\ \vdots \\ r_k \\ \vdots \\ r_m \end{pmatrix} \quad y = \begin{pmatrix} 0 \\ \vdots \\ y_k \\ \vdots \\ 0 \end{pmatrix} \quad (r - y) = \begin{pmatrix} r_1 \\ \vdots \\ r_k - y_k \\ \vdots \\ r_m \end{pmatrix}$$

where $(r - y)$ is the portfolio after trading y_k shares of stock k . Notice that the y vector only contains one value y_k for the stock that we are looking to trade. Having zeros in the other entries ensures that the other position sizes remain constant. Our goal is to determine the value of y_k that will minimize portfolio risk.

Portfolio risk after trading will be:

$$Risk = \sqrt{(r - y)'C(r - y)}$$

The number of shares to trade that will minimize total portfolio risk is determined by differentiating portfolio risk with respect to y_k , setting this derivative equal to zero, and solving for y_k . Mathematically, this is:

$$\frac{\partial Risk}{\partial y_k} \sqrt{(r - y)'C(r - y)} = 0$$

Solving, we get:

$$y_k = \frac{1}{\sigma_k^2} \sum_{j=1}^n r_j \sigma_{ij} \quad (9.27)$$

If an investor needs to adhere to the shrinking portfolio constraint, the feasible trading interval are values between zero and the original position size r_k . Recall that we are using a one-sided portfolio formulation and we have already adjusted the covariance matrix. This constraint is:

$$0 \leq y_k \leq r_k$$

Thus the actual number of shares that can be traded adhering to the shrinking portfolio constraint is:

$$y_k^* = \min(\max(0, y_k), r_k)$$